

Tracking Natural Disasters from Space:

An Exploratory Data Analysis of NASA EONET Events and FIRMS Fire Detections

I. Introduction

1. Project Aims

Human beings are often reminded of how small they are in the face of nature. Natural hazards such as wildfires, volcanic eruptions have long been a source of destruction. Although modern technology has improved our ability to observe the Earth, many natural disasters still occur suddenly and remain difficult to predict with full accuracy. Therefore, it is important to study natural hazard data in a systematic way.

The aim of this project is to **collect and analyze historical data on natural hazard events** in order to better understand where these events occur, how frequently they happen, and what patterns they show over time and across regions. By identifying the characteristics of disaster-prone areas, this project hopes to provide a clearer picture of global hazard activity and support future efforts in disaster preparedness, risk awareness, and environmental monitoring.

2. Data Sources: NASA EONET and NASA FIRMS

This project uses two public data platforms from NASA: the **Earth Observatory Natural Event Tracker (EONET)** and the **Fire Information for Resource Management System (FIRMS)**. Both are supported within NASA's Earth observation data system and offer scientific reliability. EONET is an open-source API that provides curated, continuously updated, near real-time metadata on natural events around the world, while FIRMS delivers near real-time active fire and thermal anomaly data based on satellite observations.

EONET focuses on natural hazard events at the event level. It brings together information on categories such as **wildfires, volcanoes, severe storms** and other large-scale environmental events. FIRMS, by contrast, is more specialized and detailed: it uses **MODIS** and **VIIRS** satellite observations to detect active fires and thermal anomalies, providing hotspot-level fire records with much finer spatial and temporal detail.

II. Main Data Analysis

1. Get ready the main dataset: EONET

For the main data, we obtain natural hazard event data using the NASA EONET API, which provides continuously updated information on global natural events. The returned data contains a list of events, each with several key attributes. By inspecting the dataset, we get 8 variables from the raw data and split them into 17 features for further analysis, for example:

- **Id**: unique identifier
- **Title**: event name
- **Categories**: types of natural hazards, such as wildfire or volcano
- **geometry**: featured by **coordinates** and **timestamps**

- **magnitude**: intensity of the events featured by **value and unit**
- ...

Among these variables, the **event categories** serve as the **primary target variable**, as the project focuses on identifying patterns in specific hazard types—particularly wildfires and volcanic eruptions—across geographic locations, time, and intensity indicators such as magnitude.

1.1 Obtain Data from NASA-EONET

1.2 Identify and narrow down the Target

1.3 Build up the Main Data Frame

1 to 25 of 6892 entries Filter ?

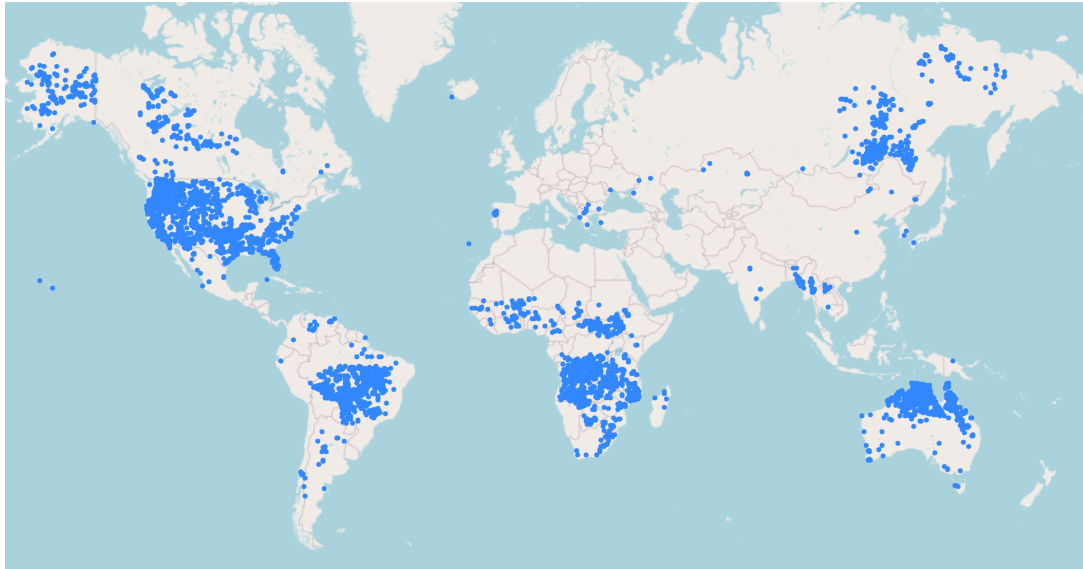
index	event_id	title	closed	category_id	category_title	source_count	geometry_date	geometry_type	magnitude_value	magnitude_unit	magnitude_description	coordinates	longitude
0	EONET_18698	Tropical Cyclone Narelle	null	severeStorms	Severe Storms	1	2026-03-17 00:00:00+00:00	Point	40.0	kts	null	156.6,-12.3	156.6
1	EONET_18698	Tropical Cyclone Narelle	null	severeStorms	Severe Storms	1	2026-03-17 06:00:00+00:00	Point	55.0	kts	null	156.2,-12.5	156.2
2	EONET_18698	Tropical Cyclone Narelle	null	severeStorms	Severe Storms	1	2026-03-17 12:00:00+00:00	Point	60.0	kts	null	155.3,-12.4	155.3
3	EONET_18698	Tropical Cyclone Narelle	null	severeStorms	Severe Storms	1	2026-03-17 18:00:00+00:00	Point	65.0	kts	null	154.3,-12.3	154.3
4	EONET_18698	Tropical Cyclone Narelle	null	severeStorms	Severe Storms	1	2026-03-18 00:00:00+00:00	Point	65.0	kts	null	153.3,-12.4	153.3
5	EONET_18698	Tropical Cyclone Narelle	null	severeStorms	Severe Storms	1	2026-03-18 06:00:00+00:00	Point	80.0	kts	null	151.9,-12.7	151.9
6	EONET_18698	Tropical Cyclone Narelle	null	severeStorms	Severe Storms	1	2026-03-18 12:00:00+00:00	Point	90.0	kts	null	150.6,-13	150.6
7	EONET_18698	Tropical Cyclone Narelle	null	severeStorms	Severe Storms	1	2026-03-18 18:00:00+00:00	Point	100.0	kts	null	149.3,-13.3	149.3
8	EONET_18698	Tropical Cyclone Narelle	null	severeStorms	Severe Storms	1	2026-03-19 00:00:00+00:00	Point	115.0	kts	null	147.8,-13.6	147.8
9	EONET_18698	Tropical Cyclone Narelle	null	severeStorms	Severe Storms	1	2026-03-19 06:00:00+00:00	Point	125.0	kts	null	146.5,-13.6	146.5
10	EONET_18698	Tropical Cyclone Narelle	null	severeStorms	Severe Storms	1	2026-03-19 12:00:00+00:00	Point	125.0	kts	null	145.4,-13.6	145.4
11	EONET_18698	Tropical Cyclone Narelle	null	severeStorms	Severe Storms	1	2026-03-19 18:00:00+00:00	Point	120.0	kts	null	144,-13.8	144.0
12	EONET_18698	Tropical Cyclone Narelle	null	severeStorms	Severe Storms	1	2026-03-20 00:00:00+00:00	Point	105.0	kts	null	143,-13.4	143.0
13	EONET_18698	Tropical Cyclone Narelle	null	severeStorms	Severe Storms	1	2026-03-20 06:00:00+00:00	Point	100.0	kts	null	142,-13.4	142.0
14	EONET_18647	MICAWBER Wildfire, Creek	null	wildfires	Wildfires	1	2026-03-16 09:20:00+00:00	Point	673.0	acres	null	-96.503611,35.664722	-96.503611

*Due to the display limit, this screenshot is just part of the full table

2. Analysis of main data

2.1 Wildfire Spatial Distribution (Visualize Target vs. coordination)

The densest concentrations appear in **North America** (especially the United States and western Canada), **the Amazon basin** in South America, **central and southern Africa**, **eastern Siberia**, and **Australia**.



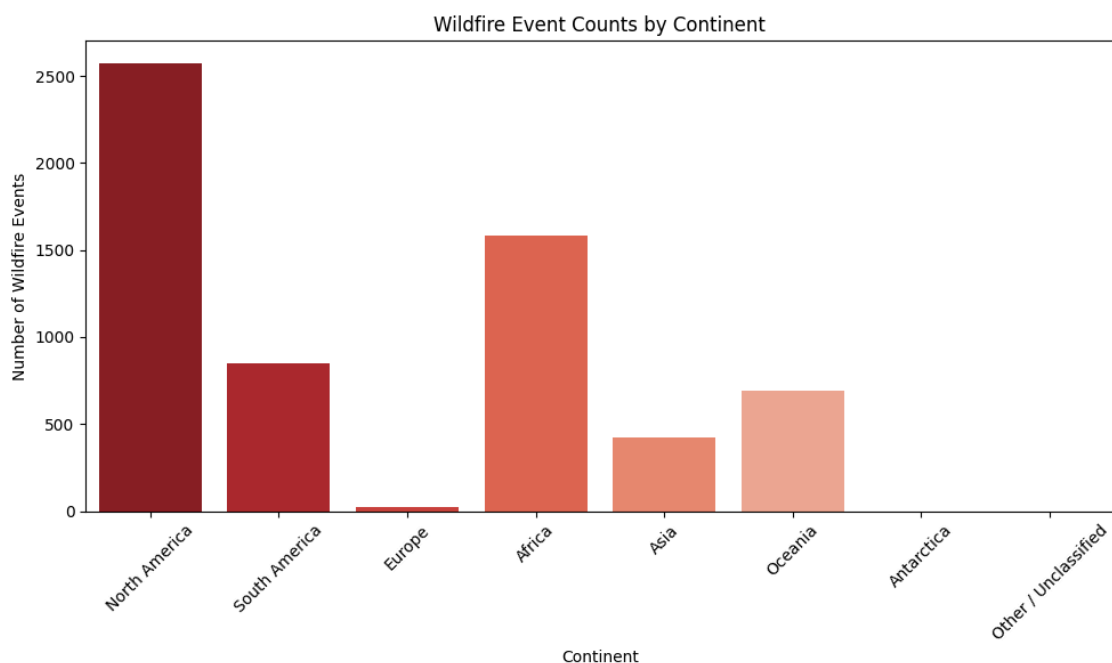
The clustering also highlights several global wildfire “hotspots,” indicating that wildfire risk is **concentrated in specific ecological regions** rather than evenly distributed worldwide.

Overall, the visualization suggests that wildfire activity is strongly associated with **vegetation availability and regional climate patterns**, particularly in tropical forests, savannas, and temperate forest zones.

2.2 Wildfire Count by Continent (Target vs. Continent)

The highest wildfire event counts appear in **North America**, followed by **Africa**, with substantial activity also recorded in **South America** and **Oceania**. In contrast, **Asia** shows a lower count, while **Europe** contains comparatively few wildfire events in this dataset.

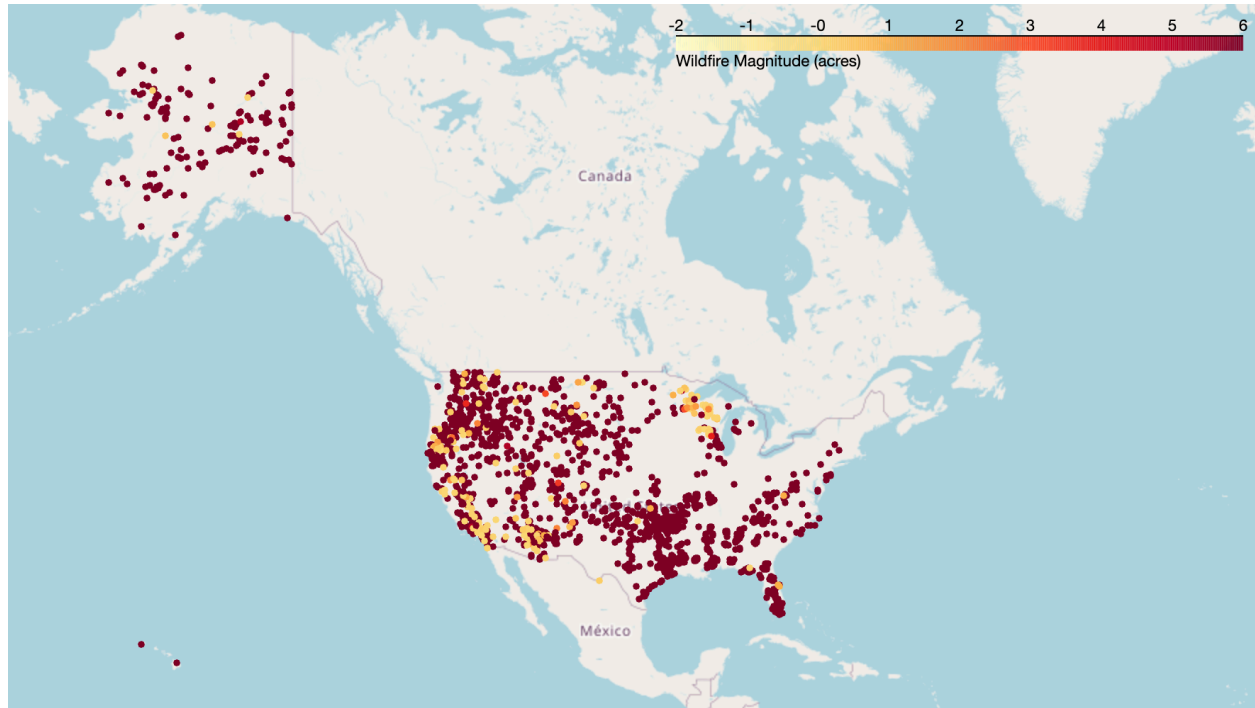
*Notes: Since there is no formal continent classification directly defined by latitude and longitude, this study assigns continents using **approximate latitude–longitude** bounding boxes. Records outside these predefined geographic ranges are labeled as **Other / Unclassified**.



2.3 Wildfire Magnitude Distribution (Target vs. Magnitude)

When examining **available magnitude data**, the majority of usable observations are concentrated in the **United States**. The map shows that most wildfire events with recorded magnitudes cluster across the **western and southern United States**, with particularly dense activity along the **West Coast and interior regions**.

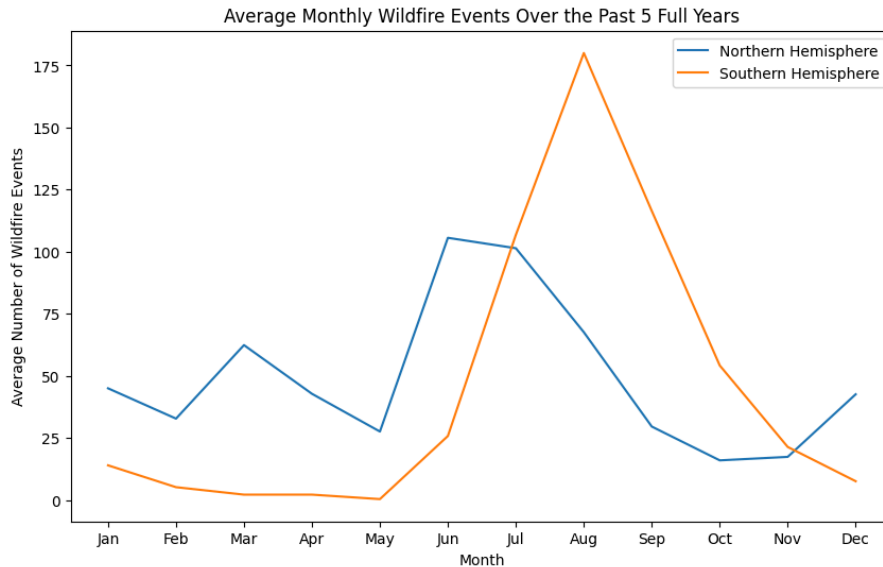
The visualization highlights the **regional variation** in wildfire size across the United States.



2.4 Monthly Wildfire Seasonality by Hemisphere (Target vs Time & Hemisphere)

Over the past several full years, wildfire activity shows a clear **seasonal pattern** that differs across hemispheres. In the **Northern Hemisphere**, wildfire events tend to **rise during the middle of the year and peak in the warmer months**, while the **Southern Hemisphere** shows relatively higher activity later in the calendar year, reflecting its **opposite seasonal cycle**. This contrast suggests that wildfire occurrence is strongly influenced by climate and seasonal dryness rather than being evenly distributed throughout the year.

The comparison highlights that global wildfire patterns are **seasonal**, but the timing of peak activity **reverses between the two hemispheres**.



III. Secondary Analysis: Using FIRMS Wildfire to qualify EONET results

The **VIIRS_SNPP_NRT** dataset from FIRMS provides near-real-time satellite detections of active fires collected by the VIIRS sensor aboard the Suomi NPP satellite, including information on fire **location**, detection **day/night time**, and **fire radiative power (FRP)**, which measures fire **intensity**.

Because EONET captures only large, discrete wildfire events while FIRMS records all fire detections (including small and short-lived fires), EONET data tends to show more **concentrated patterns across states and seasonal spikes by month**, whereas FIRMS produces **smoother, more widespread distributions** across both state-level and monthly analyses.

1. Do wildfires happen more during the day or at night?

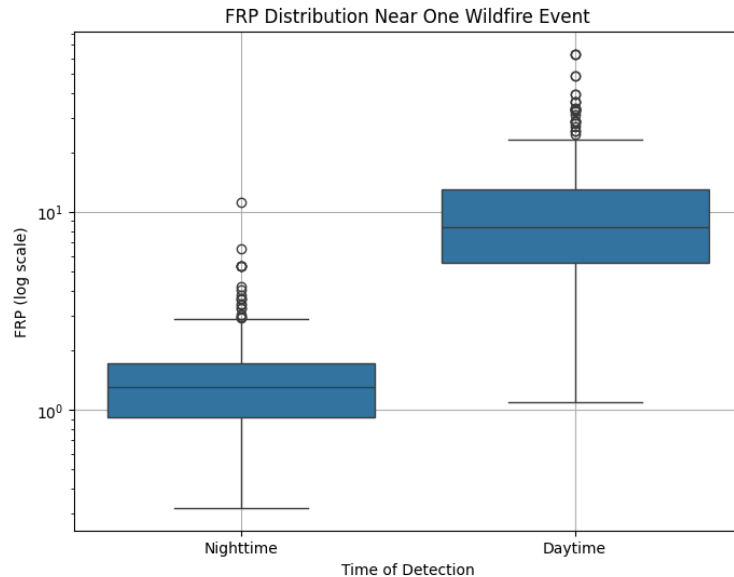
1.1 Finding based on EONET

Wildfire events in the EONET dataset occur **more frequently during the daytime (61%)** than at night (39%), suggesting that recorded wildfire activity is more commonly associated with daytime conditions or daytime satellite detection.

1.2 Finding based on FIRMS

The FIRMS data shows a strong imbalance between **daytime (87%)** and **nighttime (13%)** fire detections, suggesting that **wildfire activity in this area is much more frequently detected under daytime conditions**. However, this pattern may reflect not only actual fire behavior but also differences in satellite detection capabilities, as daytime observations can capture larger or more visible fires more effectively.

*Notes: Since the fire spots offered by FIRMS are not necessarily wildfires, the study used the location information from EONET to capture events from FIRMS to align the event type.

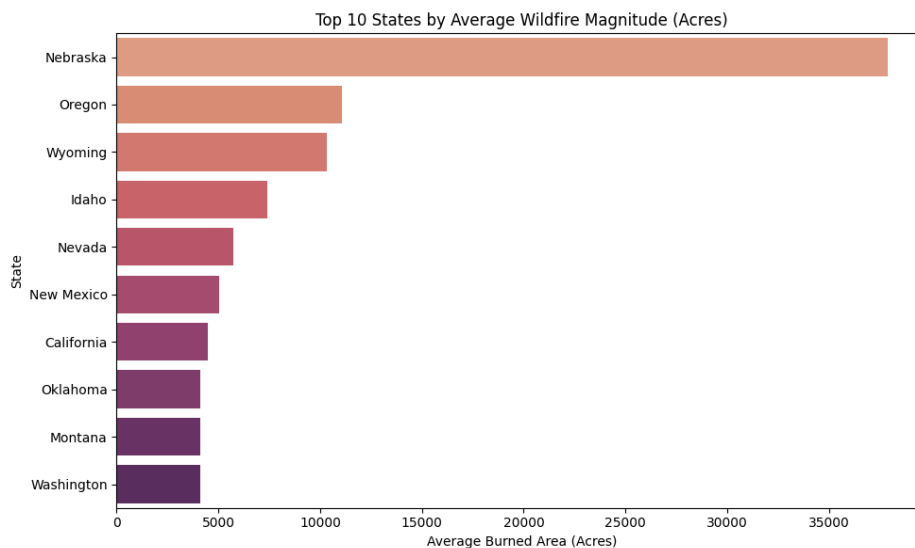


2. How does the Fire Power Radiation vary inside the United States?

The availability of wildfire magnitude data in the EONET dataset is highly **uneven across regions**, with most magnitude measurements concentrated in North America. In response to this data feature, the study zooms in to see the fire intensity variation inside the US by comparing magnitude **value** from EONET and **FPR** from FIRMS.

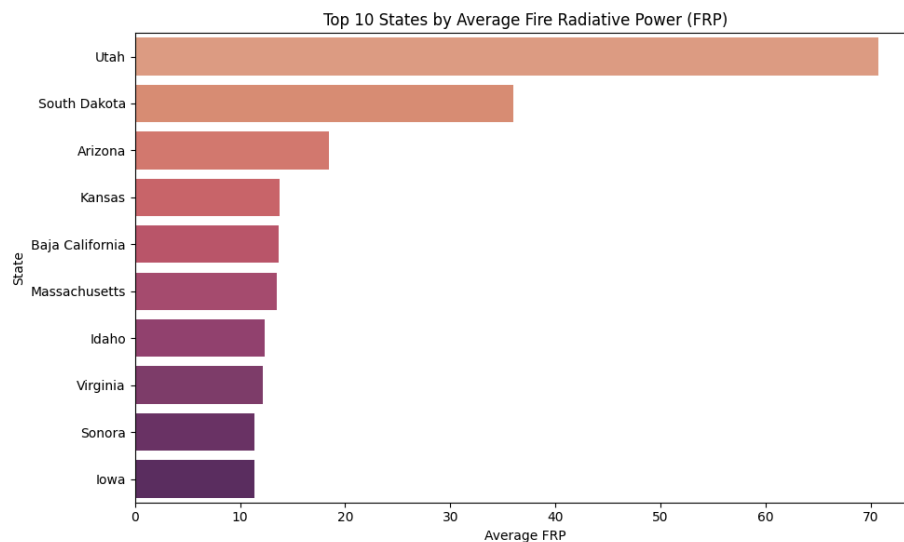
2.1 Finding based on EONET (Wildfire event by Magnitude)

When focusing on U.S. data, clear regional patterns emerge: **western states** consistently exhibit **larger average burned areas**, indicating that wildfires in these regions tend to spread over greater extents. The results show that a small number of states—particularly **Nebraska** and several western states have significantly higher average wildfire magnitudes than others, indicating a **highly skewed distribution** of wildfire size across the United States.



2.2 Finding based on FIRMS (Fire intensity by FPR)

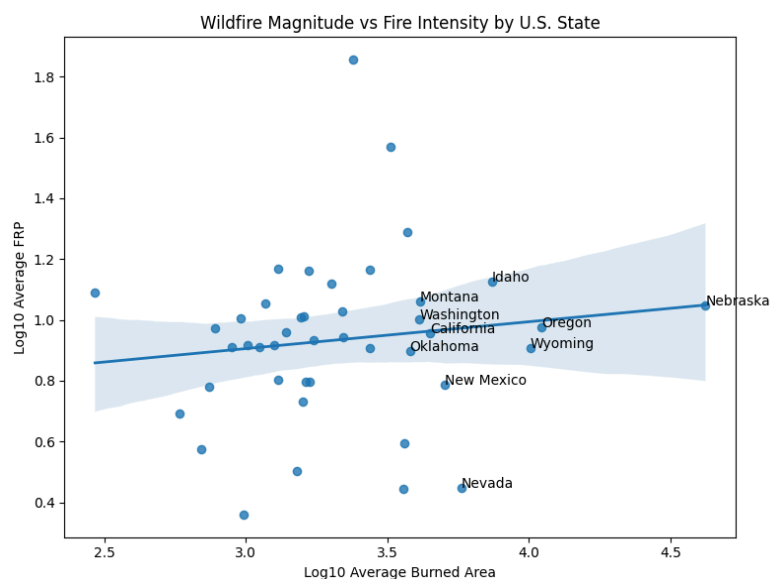
The results indicate that wildfire intensity, as measured by average FRP, varies notably across states, with **Utah** exhibiting substantially higher fire intensity than others, with an average FRP of about 70.74. Overall, the distribution suggests that fire intensity is **unevenly distributed**.



2.3 Associate EONET and FIRMS (Magnitude vs. Intensity)

By combining EONET and FIRMS, a more comprehensive picture of wildfire behavior emerges. Their comparison reveals a generally consistent pattern: **regions with larger average burned areas often also exhibit higher fire intensity**. This relationship is particularly evident in western U.S. states, where both magnitude and FRP are elevated.

However, the relationship is not perfectly linear, reflecting differences in fire behavior, environmental conditions, and data structure between the two sources.

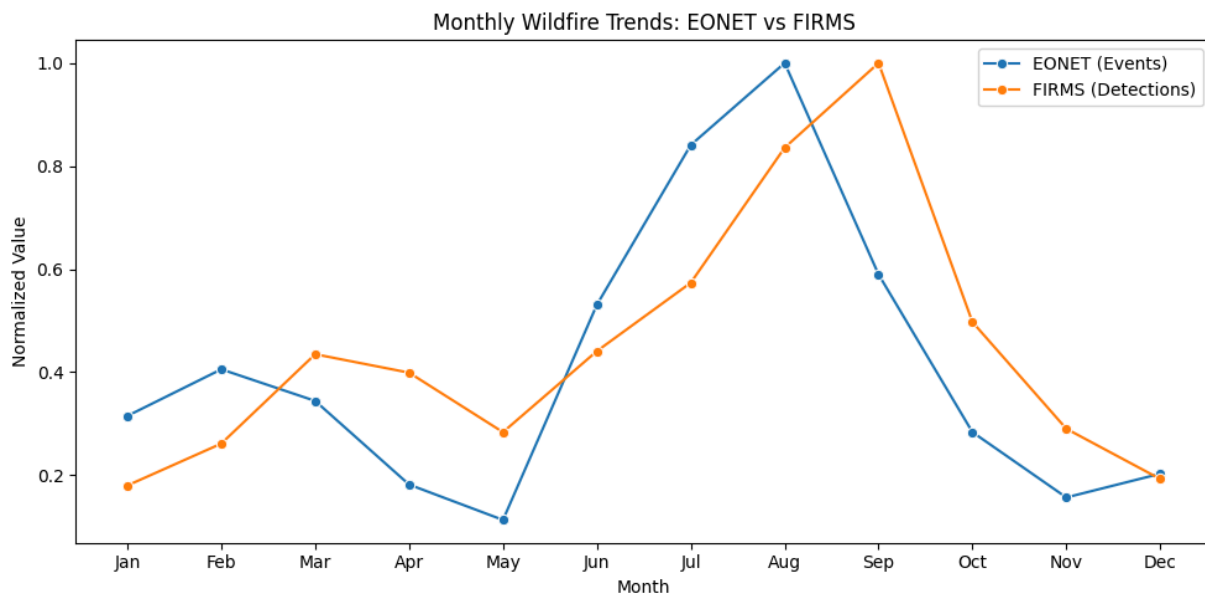


3. What is the seasonal pattern of wildfire?

Wildfire activity in the United States is **strongly seasonal**, with **late summer and early fall** representing the most active and intense period. The consistency between EONET and FIRMS reinforces the reliability of this pattern, while the addition of FRP highlights that **seasonal factors influence both the occurrence and intensity** of wildfires.

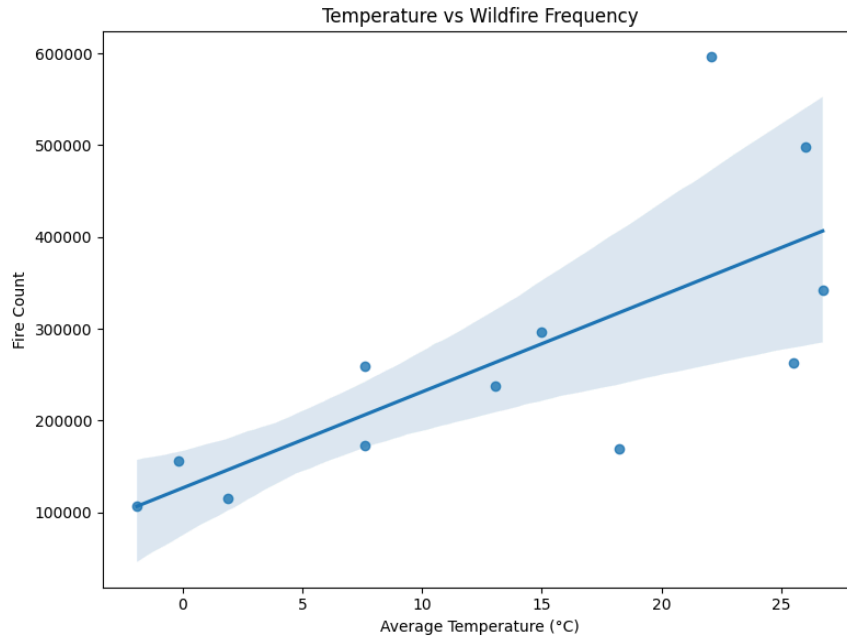
3.1 Combine EONET and FIRMS

Both EONET and FIRMS datasets exhibit **similar seasonal patterns**, with wildfire activity peaking during summer months. This consistency suggests that EONET event reporting aligns well with satellite-based fire detections. Their strong alignment suggests that **both frequency and intensity of wildfires are driven by similar seasonal factors**.



3.2 Seasonal Factor: Temperature

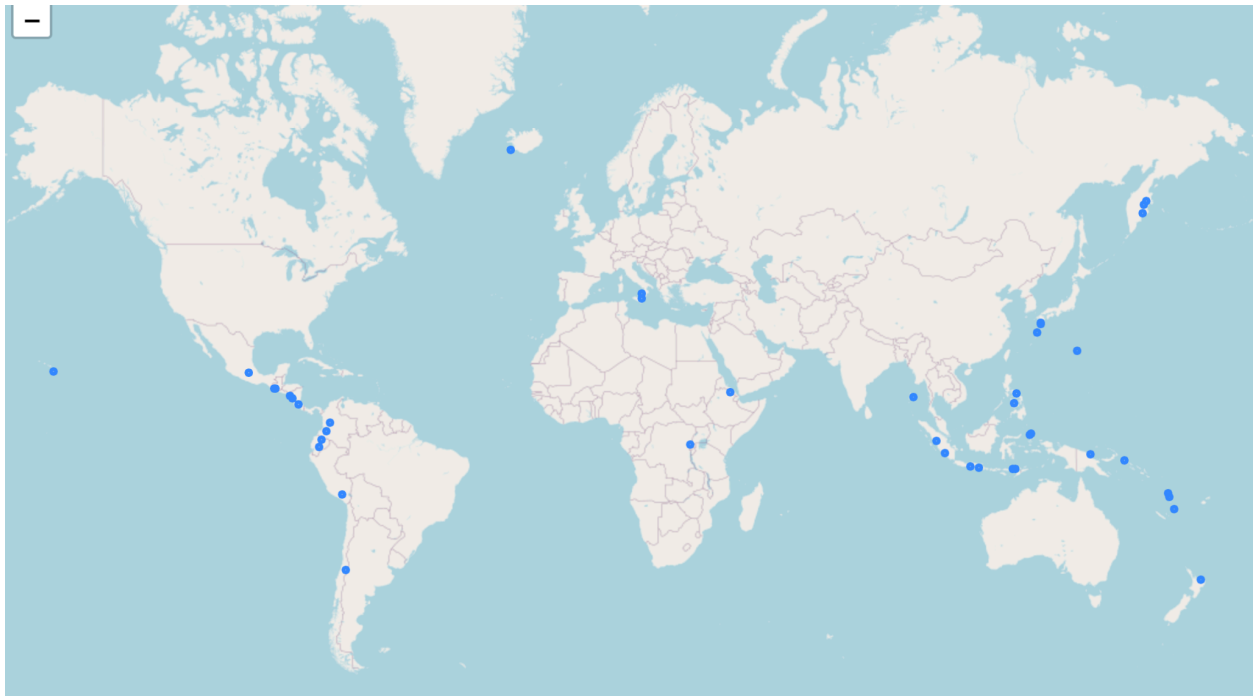
The results show a clear **seasonal temperature pattern**, with temperatures rising from winter to a peak in July–August and then declining toward the end of the year. This pattern closely aligns with wildfire activity, as indicated by the **strong positive correlations between temperature and both fire frequency (0.73) and fire intensity (FRP, 0.81)**. The scatter plot further confirms this relationship, showing that **higher temperatures are associated with substantially higher wildfire counts**. Overall, the findings suggest that temperature is a key driver of both the occurrence and severity of wildfires, reinforcing the idea that seasonal climatic conditions play a crucial role in shaping wildfire dynamics.



4. Additional Analysis: Volcanic Activities

4.1 Volcanic Activities Spatial Distribution

Volcanic activities are mainly concentrated along the **Pacific Ring of Fire**, especially the **western coasts of North and South America, East and Southeast Asia, and Oceania**. Additional smaller clusters appear in the **Mediterranean region, Iceland, and East Africa**.



4.2 Volcanic Seasonality by Hemisphere

Similar to wildfire, volcanic activity also demonstrates a clear **seasonal pattern** that varies across hemispheres. In the **Northern Hemisphere**, volcanic activities are more likely to happen from **June to August**, which is known as **summer**. While for the **Southern Hemisphere**, volcanic activities take place frequently from **November to January**, reflecting their **opposite seasonal cycle**. This contrast suggests that for volcanic activities, their occurrence is also strongly influenced by climate and seasonal dryness.

Overall, the comparison highlights that global volcanic activity patterns are **seasonal**.

hemisphere	month	month_name	Northern Hemisphere	Southern Hemisphere
0	1	Jan	3.0	3.0
1	2	Feb	0.0	1.0
2	3	Mar	1.0	1.0
3	4	Apr	2.0	0.0
4	5	May	NaN	NaN
5	6	Jun	4.0	0.0
6	7	Jul	3.0	1.0
7	8	Aug	5.0	3.0
8	9	Sep	2.0	0.0
9	10	Oct	1.0	0.0
10	11	Nov	3.0	2.0
11	12	Dec	2.0	7.0

IV. Limitations and Possible Improvements

This project has several limitations. First, the EONET dataset has a limited sample size and category richness. Compared with **wildfires**, other categories, such as **volcanoes** and **sea and lake ice**, contain far fewer records, so the analysis cannot be equally extended to all natural hazards.

Second, the FIRMS data used in this study are mostly concentrated in **North America**, resulting in less strong global generalizability. Furthermore, FIRMS records **fire detections** rather than independent wildfire events. A single large wildfire may produce many fire points, and some detected points may just represent small-scale burning.

Finally, this project matches FIRMS and EONET wildfire records mainly through approximate **latitude-longitude ranges**. The method is not fully precise and may cause missing or repeated records.

V. Conclusion

This project finds that wildfire and volcanic activity follow clear **spatial and seasonal** patterns, with wildfire events concentrated in specific regions globally and highly uneven in magnitude and intensity across U.S. states. By combining EONET event data with FIRMS satellite detections, the wildfire analysis shows **strong consistency between the two sources**, including a clear dominance of daytime detections and peak wildfire activity in late summer and early fall. Furthermore, **temperature** is strongly positively correlated with both wildfire frequency and intensity, indicating that seasonal climatic conditions are a key driver of wildfire behavior and helping explain when and where wildfires are most severe.

References:

1. NASA. (n.d.). *Earth Observatory Natural Event Tracker (EONET)*. NASA. <https://eonet.gsfc.nasa.gov/>
2. NASA. (n.d.-b). *Firms*. NASA. <https://firms.modaps.eosdis.nasa.gov/>
3. *Free weather API*. Open. (n.d.). <https://open-meteo.com/>